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The effect of toolpath design on the mechanical strength of additively manufactured glass components

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Philipp Amir Chhadeh[1,3], Emanuel Nowak[1,3], Dylan Vlahopoulos[2,3] ✉, Matthias Seel[1] & Nick Birbilis[2] ✉

To date, there has been a paucity of results regarding mechanical properties of additively manufactured (AM) glass. The findings herein provide valuable insights into optimising AM processes for glass, aiming to enhance the reliability, structural performance, and practical application of glass components in both engineering and architectural contexts. This study investigates the mechanical behaviour of glass components fabricated using a Fused Deposition Modelling (FDM) type technique. The research focuses on understanding the bending strength and overall structural integrity of FDM-manufactured glass, in comparison to conventionally prepared glass, with particular attention to the layer-to-layer bonding imparted by the AM process. Soda-lime glass rods, produced via an automated pulling method, were used as filament feedstock for the AM process. Mechanical testing, including four-point bending, highlighted the influence of layer orientation and AM toolpath on mechanical performance. The results show toolpath optimisation can improve strength by controlling heat retention during deposition.

Glass has long been a material of choice in numerous engineering applications due to its transparency, durability, and resistance to environmental factors. Traditional glass manufacturing methods, such as float glass production and casting, have been extensively studied and optimised. The float process is the standard industrial process for flat glass production for more than 60 years. The material properties of glass, such as strength, are therefore known and accepted, including in standardisation (see EN 572-1[1]). However, these methods often face limitations when it comes to producing complex shapes. The recent advent of additive manufacturing (AM) techniques offers a potential solution to these challenges, allowing for greater design freedom and the ability to tailor mechanical properties.

The direct AM (i.e., '3D printing') of glass presents significant challenges due to the high melting temperatures required, the non- linear thermomechanical properties, the development of undesired residual stress during the cool process, and the inherent brittleness of the material. Additionally, the low process tolerance and the necessity for achieving full density further complicate the AM process. Even if the field of AM glass is difficult and new, various AM glass techniques have become established in recent years. The four most common are briefly explained here. Selective laser melting (SLM) uses a high-power laser to fuse glass powder layer by layer, resulting in precise and intricate, but also opaque structures[2,3]. Fused deposition modelling (FDM), also known as Fused filament facrication (FFF), involves extruding molten glass through a nozzle, which solidifies upon deposition, making it suitable for larger forms, as depicted in Fig. 1[4–6]. Stereolithography (SLA) uses a laser to cure a photosensitive resin mixed with glass particles, creating high-resolution and complex shapes[7–9]. Lastly, Directed energy deposition (DED) involves feeding glass in the form of powder or wire into a focused energy source, such as a laser or electron beam, allowing for the creation of large and robust components[10–13].

Specifically in regards to the present study, FDM, a process that involves the layer-by-layer deposition of a semi-solid material to create a three-dimensional object, is still a novel process with the material glass. This method provides several advantages, including precise control over the geometry and the ability to fabricate intricate structures. Despite these advantages, the mechanical performance of FDM-fabricated glass components is not known, particularly concerning the integrity of layer-to-layer connections. The reasons for concern include an inhomogeneous joining zone and sharp edge between the individual layers as well as the resulting residual stress at uneven process temperatures.

The objective of the present study is to characterise the mechanical behaviour of AM glass components produced via the FDM process. Understanding the mechanical properties of these components is crucial for their potential application in engineering and architectural contexts, where reliability and structural integrity are paramount. To achieve this, a series of

[1]Glass Competence Center, Institute of Structural Mechanics and Design, Technical University of Darmstadt, Darmstadt, Germany. [2]Deakin University, Faculty of Science, Engineering and Built Environment, Melbourne, VIC, Australia. [3]These authors contributed equally: Philipp Amir Chhadeh, Emanuel Nowak, Dylan Vlahopoulos. ✉e-mail: d.vlahopoulos@deakin.edu.au; nick.birbilis@deakin.edu.au

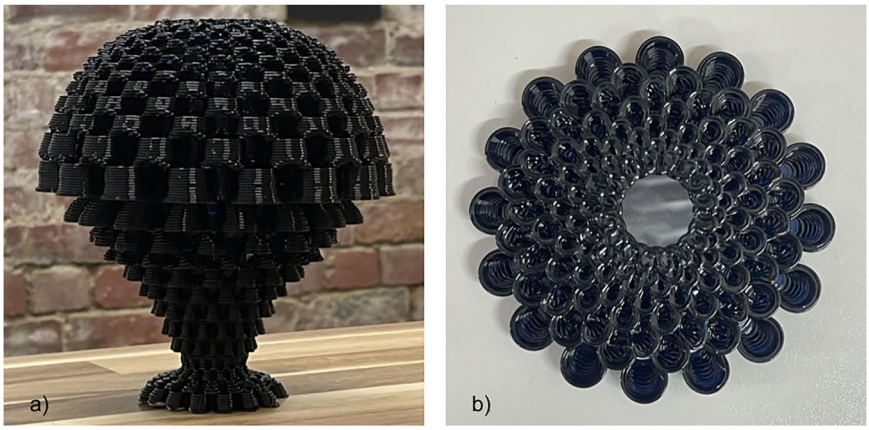


Fig. 1 | Additive manufactured glass lamp prepared by FDM (using Maple 4TM printer). a Final product assembled from three printed parts; **b** standalone part before assembly. Material: Kugler glass (K047 aquamarine blue).

Table 1 | Machines and components used for the four-point bending test

Component	Description
Testing Machine	ZwickRoell Z050
Load Cell	ZwickRoell Performance P (within the range of 5 N-150 kN linearity of ± 1%)
Displacement Transducer	Laser displacement sensor Keyence IL-030 (in the range of 20-45mm linearity of ± 0.1%).
Rollers	Made of stainless steel with a radius of 6 mm, length 100 mm
Elastomer on Rollers	3 mm thick EPDM conforming to ISO 48 (IRHD: 40 ± 10)
Testing Software	ZwickRoell testXpert III

mechanical tests are conducted to examine the strength of the layer-to-layer connections in FDM-manufactured glass specimens as a function of the layer arrangement and orientation.

The advancement of AM technologies has potential to revolutionise glass fabrication - finally enabling reproducible, intricate and customised glass components with high precision (and without the need for human intervention or artisanal skills). This may be realised by either standalone (final net-shape) AM glass components or AM glass components fused to either existing glass products or other AM prepared components.

AM glass has emerged as a tool to simplify material manipulation through the use of digital designs and automation. The ability to expedite the feedback loop with glass materials, from concept to creation, holds immense potential for various industries. The industries in which AM glass is currently active include art and design, architecture, education, research and optics. In each of these industries, traditional glass production requires years of glass working experience or outsourcing to factories. AM glass offers the capability to rapidly manufacture glass components onsite. This is particularly beneficial to scenarios where delivery is not feasible (e.g., space or submarine environments) or to reduce reliance on supply chains.

In the case of art and design, AM glass enables the creation of unique geometries not possible with traditional glass manufacturing techniques. Internal structures, assemblies, thin features and interlocking parts are more easily created without the need for skilled artistry (e.g., glassblowing or lampworking) or additional tooling costs (e.g., moulds). This allows for bespoke designs in interior design and commercial use, including storage containers, lighting pieces, furniture and jewellery. See for example Fig. 1.

For research and development, this study exemplified how AM glass enabled mechanical testing to commence quickly, without the need for a mould or intensive manual labour. In optical applications, there is ongoing interest in using AM glass for optical fibre and lens applications. Pan[14] investigated the AM of glass preforms and successfully drew them into optical fibres, indicating a promising future for complex microstructured fibre fabrication.

AM glass also offers the ability to exploit the use of fibres during the printing process. The precise control of fibrous glass structures holds exciting potential. Additionally, many AM glass techniques allow the user to pause production, enabling modification during the print, such as embedding components or introducing unique materials.

In addition to the potential for the production of three-dimensional objects from glass, the possibility of fusing glass onto existing glass products presents a vast array of applications.

In the present era, designers are investigating the potential of AM. A multitude of concepts can be actualised through the use of glass as a material. Existent glass products can also be modified through the application of 3D printing technology. Flat objects are particularly amenable to this process. Printing can also be employed on glass tubes or other shapes.

The potential for printing on flat glass panes offers a means to achieve both enhanced design flexibility and load bearing capacity of the glass, see ref. 15. In ref. 16, an analytical study was conducted for the construction industry with the objective of optimising the deflection of a glass façade under load by applying stiffeners. The results of this study demonstrate that the application of stiffeners can result in material savings of up to 70% for a two sided supported glass plate. Additionally, the study highlights that significant savings can be achieved even when the placement of the stiffeners is not optimal. This provides designers with considerable design freedom, allowing for innovative solutions that may not have been previously considered. For instance, the redirection of solar radiation and sound waves could be utilised to enhance the comfort of the interior environment.

The conceivable applications in the field of architecture can be continued indefinitely, AM glass can be used to apply stiffeners but also fused fixings for flat glass, see ref. 17. It would also be possible to enhance the production of insulating glazing units by permanently sealing the units by using AM glass and additionally create the spacers for vacuum insulating glazing units.

Results

Setup

The machines and components used for this test can be found in Table 1.

The experimental setup illustrated in Fig. 2 follows the guidelines of EN 1288-3[18]. The dimensions are as follows:

- Support span L_s = 130 mm
- Load application span L_b = 70 mm
- Lever arm (distance between support and load application) L_w = 30 mm.

This setup deviates from EN 1288-3[18] in terms of dimensions and does not use load and support rollers that are freely rotatable along their longitudinal axis. The dimensions conform to DIN 843-1[19] for four-point testing

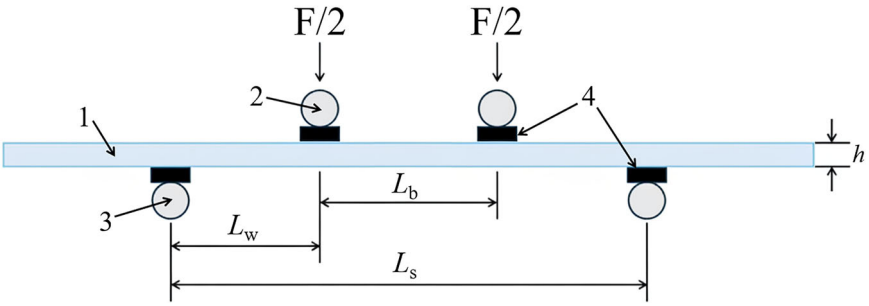


Fig. 2 | Front facing view schematic of the four-point bend test set up, where 1) test specimen 2) load roller 3) support roller 4) elastomer on rollers according to ref. 18.

of small ceramics, which specifies that the rolls must be rotatable up to 15° parallel to the longitudinal axis of the specimen. In this setup, the support rolls allow rotation of ± 10°, while the load rolls permit coupled rotation of ± 1° around the specimen's longitudinal axis (see Fig. 2). The plate with the load rolls is pivotally mounted at the point of force application, allowing the rolls to align towards the specimen. The deflection of the specimen at its centre of gravity was recorded using a laser displacement sensor.

Specimen production

A dog-bone geometry was selected as the specimen type. The 3D model was first created using computer aided design software. This was then inputted into the slicing software to prepare the files for 3D printing. All samples featured zero perimeters and 100% rectilinear infill, with various infill orientations. 'PrusaSlicer 2.6' was employed to prepare the 45° and 90° samples. However, the 0° samples were created using 'Grasshopper 3D + Rhino', to optimise the tool path. This is because the tool path generated by the slicing software was inefficient, with potential to create inconsistent results. Settings remained consistent between all samples, to ensure comparable parameters.

A challenge encountered during the dog-bone production involved achieving consistent levelling of the first layer. Manual mesh bed levelling was employed, involving the probing of six points, which were inspected by the operator before each print. Human error is likely introduced during this process, as observing deviations smaller than 1mm through the viewing window of the heated chamber is challenging. This resulted in different levels of adhesion to the printing platform for each sample. In some samples curling of the corners was observed. This effect worsened if the printing path created a shearing force on these low adhesion areas.

Efforts were undertaken to maintain consistent parameters during the production of specimens, with the exception of toolpaths, which varied significantly across series to achieve different layer orientations. These variations in toolpaths, particularly regarding thermal management, had a substantial impact on the manufactured specimens.

The orientation of 0° specifies that the layers are oriented in the direction of bending. The value of 90° indicates that the layers are oriented in a direction transverse to the direction of bending. Finally, the value of 45° represents the orientation of the layers at an angle of 45° to the direction of bending. Accordingly, the sample code follows the format 'infill angle – sample number'. A representation of the layer progressions can be found in Fig. 3.

The toolpaths are detailed as follows:

For the 90° series, the paths were short and rapidly repeated, allowing previously printed layers to retain heat. Additionally, nozzle radiation continuously warmed the contact surfaces. As an example, Fig. 4 shows a dog-bone specimen with the 90° infill pattern. In the 45° series, the paths were approximately 50 percent longer than those of the 90° series. The nozzle paused twice during the printing of the web to complete external corners, moving more slowly at the corners of layer lines compared to the 90° series. The 0° series involved extremely long paths, with the nozzle printing over distances exceeding 300 mm to reach certain contact points. Outer contours were printed first, resulting in the subsequently printed internal lines encountering cooler contact points.

Execution

The experiment 1 was conducted at a temperature of 22 °C and a relative humidity of 44%. The experiment 2 was conducted at a temperature of 24 °C and a relative humidity of 48%. Due to the very low forces encountered, the load application rate prescribed in EN 1288-3[18] was reduced from 2 MPa/s to 1 MPa/s (equivalent to ca. 1.5 N/s). Measurements were taken every 0.1 s using the testXpert III software.

Specimens were centrally placed and aligned in the setup, with the print bed side facing upwards (into the compression zone) so that the typical AM surface was in the tension zone. A preload of 3–5 N was applied to ensure the rollers aligned properly with the specimen and made uniform contact everywhere. Same applies to the testing of the filament rods. The selected alignment was negligible due to the circle shaped cross section geometry.

When the test commenced, load, specimen's deflection, and crosshead displacement were recorded until failure. Upon fracture, the testing machine rapidly retracted to prevent further loading on the break or inducing additional fractures. Figure 5a shows the specimen 90-1 after testing, as well as the other layer orientations in Fig. 5b, c.

Experiment 1: Dog-bone bending tests

Table 2 displays the fracture load (F_u) and the calculated fracture stress (σ_u) in the specimen during four-point bending tests. The fracture stress was calculated using Equation (1), where the inner lever arm, z, is taken as the distance from the centroid of the fracture area to the fracture origin. Since the fracture origin lies on the bottom surface of the specimens, z is equal to $h/2$. The bending moment M_u is derived from the four-point bending set up, see Equation (2), where $L_w = 30$ mm. The geometrical moment of inertia, I_y, was calculated using hand measurements of the actual specimen geometry, at the location of failure after testing, rather than the nominal (target) dimensions from the design. For simplicity, a rectangular section is assumed, allowing Equation (3) to be applied, where b is the measured width of the specimen and h is the measured thickness of the specimen (see Fig. 6).

$$\sigma_u = \frac{M_u}{I_y} \cdot z \tag{1}$$

$$M_u = \frac{F_u}{2} \cdot L_w \tag{2}$$

$$I_{y,rec} = \frac{b \cdot h^3}{12} \tag{3}$$

All fractures occurred as intended within the narrow section of the dog-bone geometry, consistent with the chosen design, except of one specimen (0-3) where the fracture extended from the narrow section into the broader area. Seven out of ten specimens (0-1, 0-2, 0-4, 45-1, 45-3, 45-4, 45-5) exhibited a completely straight crack perpendicular to the longitudinal axis, indicating effective interlayer adhesion that did not follow individual layer deposition directions. This straight fracture path also suggests uniform load introduction.

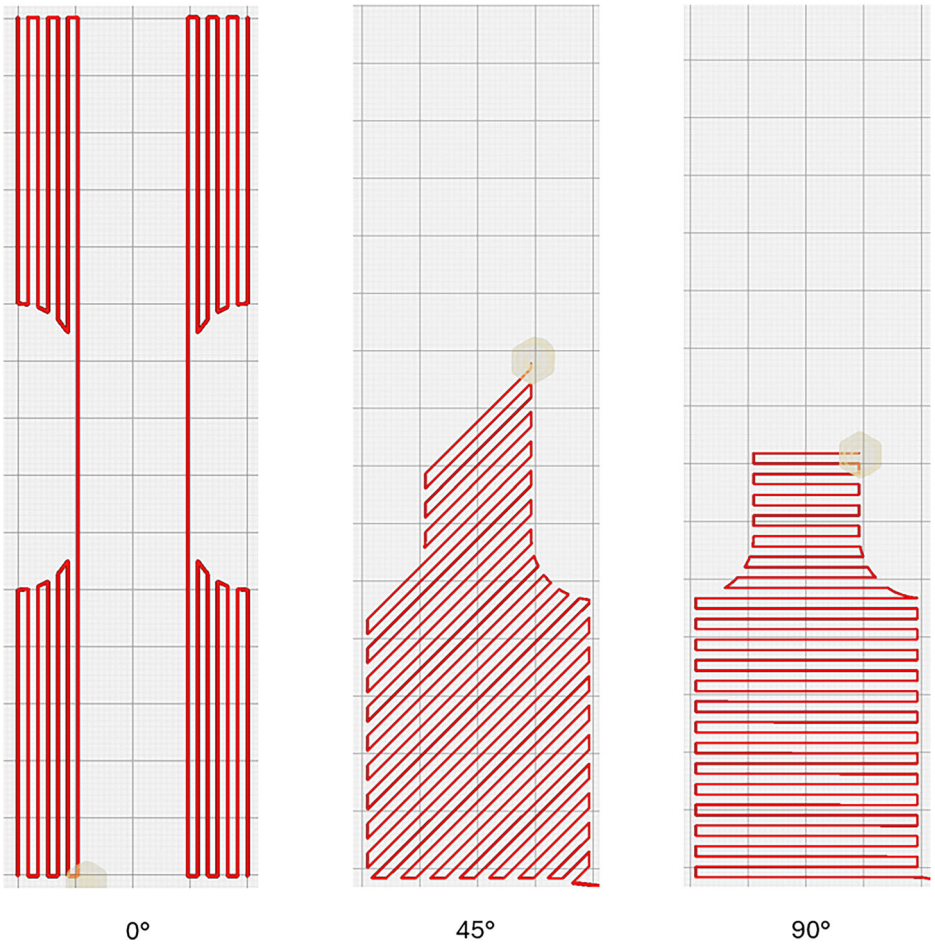


Fig. 3 | Toolpaths at 50% layer completion, for each type of specimen geometry prepared for this study.

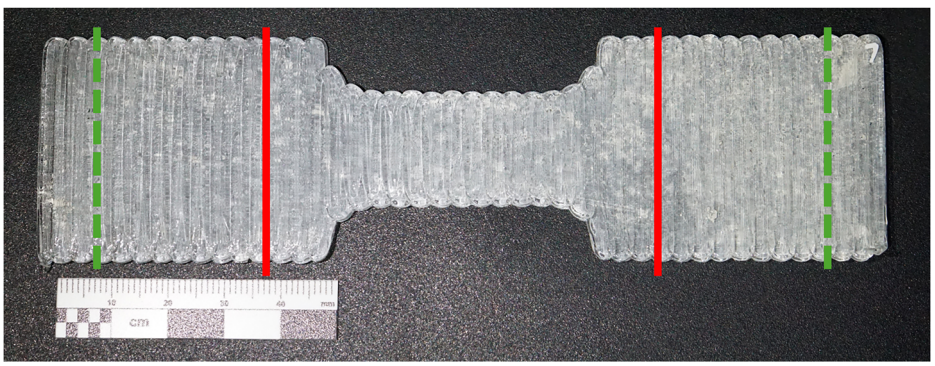


Fig. 4 | An AM glass specimen that was prepared with a 90 degree infill orientation (Specimen 90-1). The positions of the load rollers (located on the top surface) and support rollers (bottom surface) are indicated by red and green lines, respectively.

All specimens in the 90° series exhibited at least three cracks, typically originating from the same point, which did not always follow a straight path but sometimes angled through the thin web. Specimen 90-2 exhibited six cracks, indicating significant energy release upon failure, consistent with the high fracture stresses observed in the 90° series (see Fig. 5).

Experiment 2: Filament Bending

In order to put the results of the additive manufactured dog-bone geometry into perspective, the same four-point bending test was done with the 3.5 mm diameter filament rods.

For the calculation of the fracture stress of the rod, the geometrical moment of inertia for a circle was used according to Equation (4), assuming a perfectly circular cross-section for the filament rods.

$$I_{y,circ} = \frac{\pi \cdot d^4}{64} \tag{4}$$

However, the determination of the maximum inner lever arm (z) requires a discussion of the fracture origin's position. The fracture origin was not consistently located on the underside of the rod, which leads to a different distance between fracture origin and centroid of the circular cross section. Therefore, its precise position was determined for each individual specimen using a microscope and described by the angular deviation alpha. Consequently, the maximum fracture stress (σ_u) was calculated for each rod

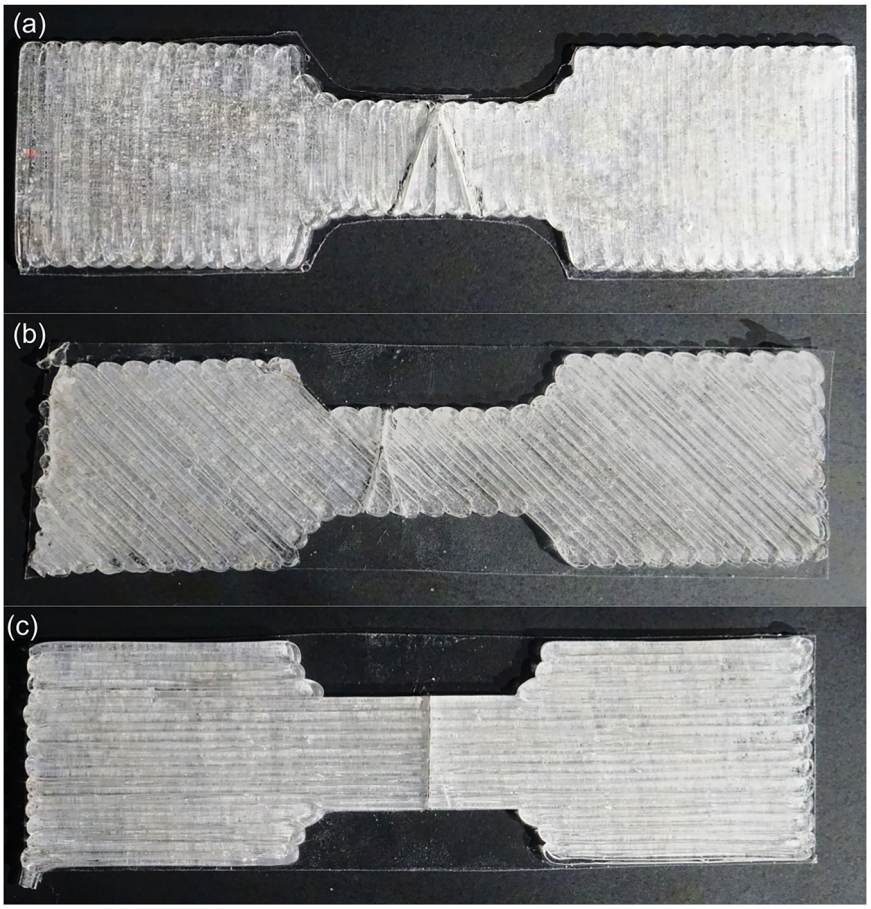


Fig. 5 | Post-testing view of AM glass specimens (in top view) type after testing defined in Table 2. Images correspond to **a** specimen 90-1, **b** specimen 45-2, **c** specimen 0-1.

Table 2 | Fracture stress σ_u of the 15 tested AM specimen under fracture load F_u from experiment 1

#	I_y mm^4	z mm	F_u N	σ_u N/mm^2
0-1	38.4	1.4	81.8	43
0-2	22.2	1.2	54.5	42
0-3	29.4	1.2	62.2	37
0-4	16.1	1.0	42.5	40
0-5	19.5	1.1	45.1	38
90-1	42.9	1.5	92.5	47
90-2	60.1	1.6	142.4	57
90-3	29.1	1.3	81.3	54
90-4	33.7	1.3	101.9	59
90-5	44.9	1.4	128.0	61
45-1	42.7	1.5	51.1	26
45-2	18.1	1.1	66.3	60
45-3	37.5	1.4	61.6	35
45-4	41.2	1.4	76.1	39
45-5	28.4	1.3	55.3	37

The values for σ_u should be considered with an approximate uncertainty of up to ± 4 N/mm^2, attributable to measurement inaccuracies in the fracture sides, the fracture load and length of lever arm.

according to Equations (1), (2), and (4), taking into account the measured diameter of the respective rod and the determined alpha.

An example is shown in Fig. 7, where a microscopic image of the rod 14 is shown in (a), the fracture origin of which is at an angle of approx. 45°. As a result, z is 1.25 mm. In (b) the deviation in rod 3 is only approx. 15° leading to a smaller inner lever arm. The inner lever arm is used to calculate the fracture stress. The individual rods, their diameters, fracture origins and the resulting fracture stress are shown in the supplementary information (see Supplementary Table 1), which gives a mean value of 73.5 N/mm^2 and a standard deviation of 12.5 N/mm^2, resulting in a coefficient of variation of 0.17. For comparison, a coefficient of variation of up to 0.3 is accepted in practice when testing float glass in accordance with EN 1288, see for reference[20].

Evaluation and Comparison

The experimental results, compared in Fig. 8, show a clear dependence on the print direction. A correlation analysis (Spearman) between the layer orientation angles (0°/45°/90°) and the fracture stress indicates that there is a statistically significant (p-value = 0.03) and a large correlation (correlation value of 0.56) according to Cohen[21]. The strong scattering of the results of the 45° specimens compared to the other directions reduces the correlation coefficient substantially.

It was initially hypothesised that the 90° layer orientation would promote premature crack propagation, due to the critical notch orientation. However, contrary to expectations, the 90° specimens exhibited the highest fracture stress (mean value 56 N/mm^2), while the 0° specimens showed the lowest (mean value 40 N/mm^2). Therefore, the orientation of the layers should be taken into account in the design of the components, in combination with the printing strategy.

One probable cause for this phenomenon is the tool path itself. In the specimens with a 90° orientation, the glass remains in a warmer state, which facilitates its effective fusion with the newly deposited glass. During the long distances the nozzle travels while producing the 0° specimens the cooling process of deposited glass impacts the layer-to-layer adhesion, leading to reduced fracture stress.

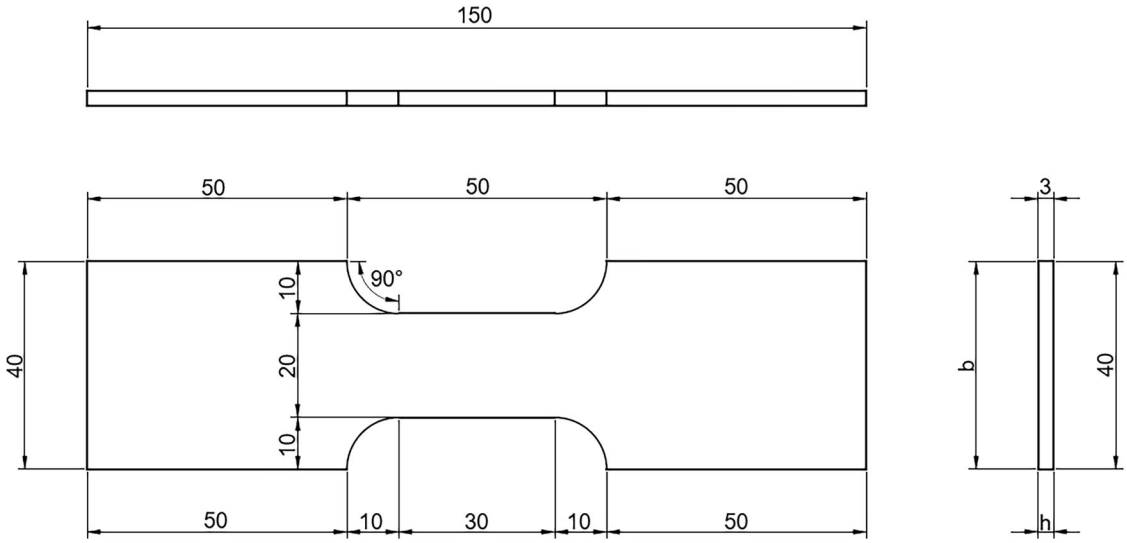


Fig. 6 | Test geometry for the dog-bone tests (measurements in [mm]; top view and two side views).

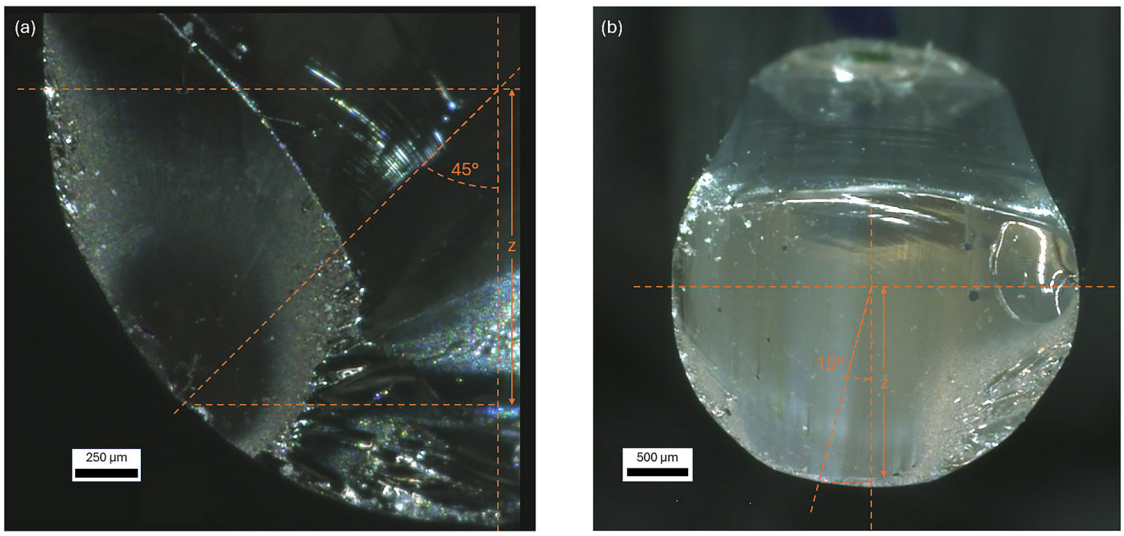


Fig. 7 | Optical microscopy images to determine the fracture origin. As per the described Experiment 2, the cross section image was used to determine the fracture origin of: **a** rod 14 with $\alpha = 45°$; **b** rod 3 with $\alpha = 15°$; in different magnifications.

Microscopic examinations of the fracture surfaces support this finding and offer a complementary explanation. For these examinations, the Dino-Light Edge Plus, was employed, utilising polarised light and multiple polarising filters in order to achieve high-contrast and informative images. The small and often non-flat fracture surface, combined with significant reflections due to the surface structure, presented a considerable challenge in terms of imaging.

The microscopy images presented include specimens from each series, with the pressurised side (also the print bed side) oriented upwards. Blue lines serve to highlight the boundaries of individual layers, a yellow circle is used to indicate the presumed origin of the fracture, and red circles are employed to mark the locations of voids. The green traces represent Wallner lines, which are visible in some specimens. The fracture mirror could not be identified using the technology employed in this study. As indicated by the red circles, a multitude of voids were observed within the fracture surfaces. A correlation was identified between the number and size of these voids and the fracture stress of the specimen. The highest-performing specimens (90° series) exhibited a paucity of voids (see Fig. 9a), whereas specimens from the 45° (see Fig. 9b) and 0° (see Fig. 9c) series displayed a considerable prevalence of such voids. It is noteworthy that the strongest specimen of the 45° series exhibited a low number of voids, which aligns with this observation. These observations were made solely on the fracture surface. To provide robust support for this hypothesis, further sectional analysis of the specimens, both longitudinally and transversely, would be necessary.

Figure 8 also shows that the rods have a higher average fracture stress than the fabricated dog-bone specimens. The decrease in this fracture stress (which for brittle materials determines the material's strength and is hereafter referred to as 'strength') with an increase in surface under tension can be described by the size effect of structural strength. The size effect on structural strength is a phenomenon observed in materials, particularly brittle materials such as glass, where the strength of a structure decreases as its size increases[22,23]. This effect is primarily due to the increased probability of encountering critical flaws or imperfections within the material as its surface area grows. The larger the structure, the more likely it is that a flaw will exist that can precipitate failure under stress.

Discussion

The test results presented in this study were analysed to characterise the mechanical behaviour of AM soda-lime glass components. The 90°

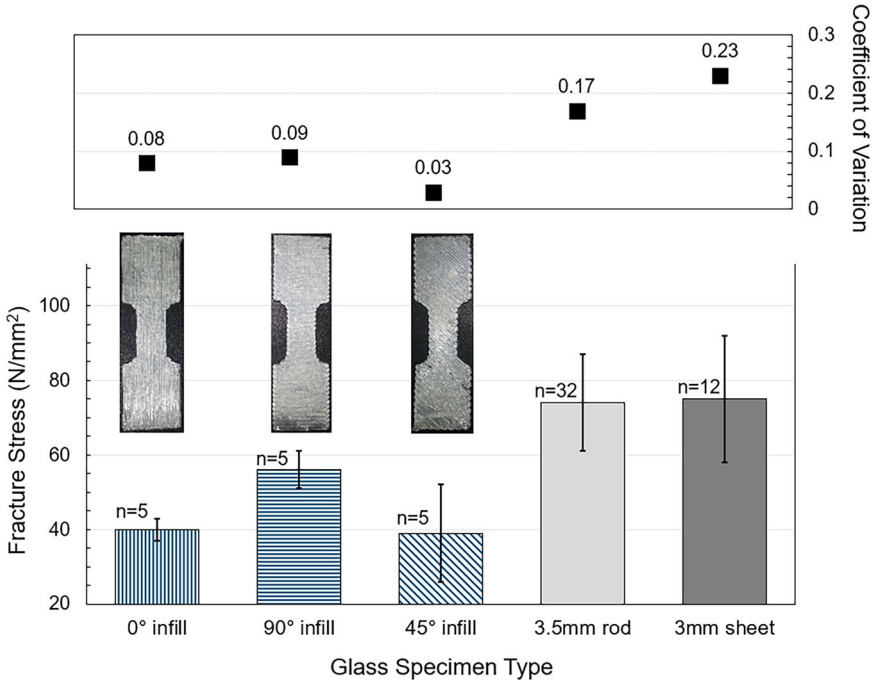


Fig. 8 | Mean fracture stress, standard deviation (error bars) and coefficient of variation for tested glass specimens. An external study with 3 mm soda-lime sheet glass is also included for comparison[26].

specimens exhibited a mean strength value approximately 40% higher than the 0° specimens. Additionally, the glass filament samples showed a 30% increase in mean strength value compared to the 90° specimens. The initial hypothesis proposed that the 0° specimens would withstand the greatest loading force before failure, with the 90° specimens expected to fail first. However, the experimental results showed the opposite. This hypothesis was based on fracture mechanics theories, which suggest that glass typically fractures at defects or discontinuities[24]. The layers in an AM component can be considered discontinuities, implying that failure is likely to occur at the interfaces between printed layers. In the 90° specimens, the infill direction forms discontinuities aligned orthogonal to the longitudinal axis, theoretically providing a low-energy fracture path. However, the disparity between theory and experimental results can be attributed to the toolpath required for each infill orientation. The 90° configuration uses a shorter, more direct toolpath with minimal travel movements, which promotes localised heating and maintains higher temperatures within the recently deposited regions. In contrast, the 0° orientation requires extended travel moves, leading to less effective heat accumulation and greater thermal dissipation. Moreover, the longer deposition paths in the 0° samples cause the tool head to alternate between recently deposited and cooler regions, potentially resulting in lower interlayer bonding strength compared to the 90° configuration. This suggests that toolpath continuity may directly influence local heat retention, which in turn affects the degree of fusion between layers. Fracture mechanics theories align with experimental results when considering the crack origin. In almost all 90° test specimens, see Fig. 5, the crack origin was commonly observed at the layer discontinuities, on the tension side. For 0° and 45° specimens, it is difficult to make conclusive statements regarding crack origin locations. Given that the fracture lines in the 90° specimens were not observed to follow the discontinuity lines, this indicates that strong interlayer bonds were successfully formed during production.

These findings indicate that AM glass specimens produced via filament extrusion exhibit a complex relationship between the thermal profiles generated by the AM toolpath and interlayer bond strength. A critical insight from this investigation is that the toolpath significantly affects heating, thereby influencing interlayer bonding strength. It is recommended that continuous and concise deposition paths be employed during printing to optimise the thermal effects from nozzle-generated radiation and convection. Furthermore, the coefficient of variation for all 3D-printed components averaged 0.16, significantly lower than the industry standard of 0.3 for sheet glass[20]. This could be an indication of the consistency and reliability of the glass AM process, particularly if a consistent toolpath (ensuring consistent heat distribution) is employed.

Future advancements in this research could explore advanced toolpaths involving non-planar movements. The results show the dependence of structural performance on the toolpath. These aspects must be further investigated to refine printing processes and to determine the optimum deposition strategy for layering, tailored to the specific requirements of each application. In the immediate term, investigations into parameters such as printing speed, layer count (part thickness), deposition size, and printing temperatures could provide valuable insights. Specifically, a detailed investigation into the formation of pores within the AM specimens should also be conducted. Increasing the overlap between layers to reduce pore size would be of interest in assessing how this influences fracture strength. Future work should increase the sample size of AM test specimens to enhance statistical analysis and further investigate variation between components. Additionally, post-processing techniques, including reheating or chemical and thermal pre-stressing, hold promise for enhancing the material properties, like strength. With this knowledge of the process and the material, an understanding of the structural behaviour of complex geometries, such as those shown in Fig. 1, can be estimated in the future. The results of this paper are an important step forward in ensuring that the visions outlined in Chapter 2 will be realised. The ability to pause AM processes offers opportunities to integrate powders, alternative materials, or objects, thereby increasing versatility. The evolution of glass AM is expected to benefit from a range of hardware and software innovations. Notable directions include adjustments to nozzle outlet shape and size and the introduction of composite materials through co-extrusion.

Methods

In this section the production of the filaments for the AM process and the AM glass specimen, as well as the applied mechanical testing methods are presented.

Filament production

In order to 3D print glass a suitable filament is needed, in this case thin glass rods. Thin glass rods may be manufactured with various techniques. An

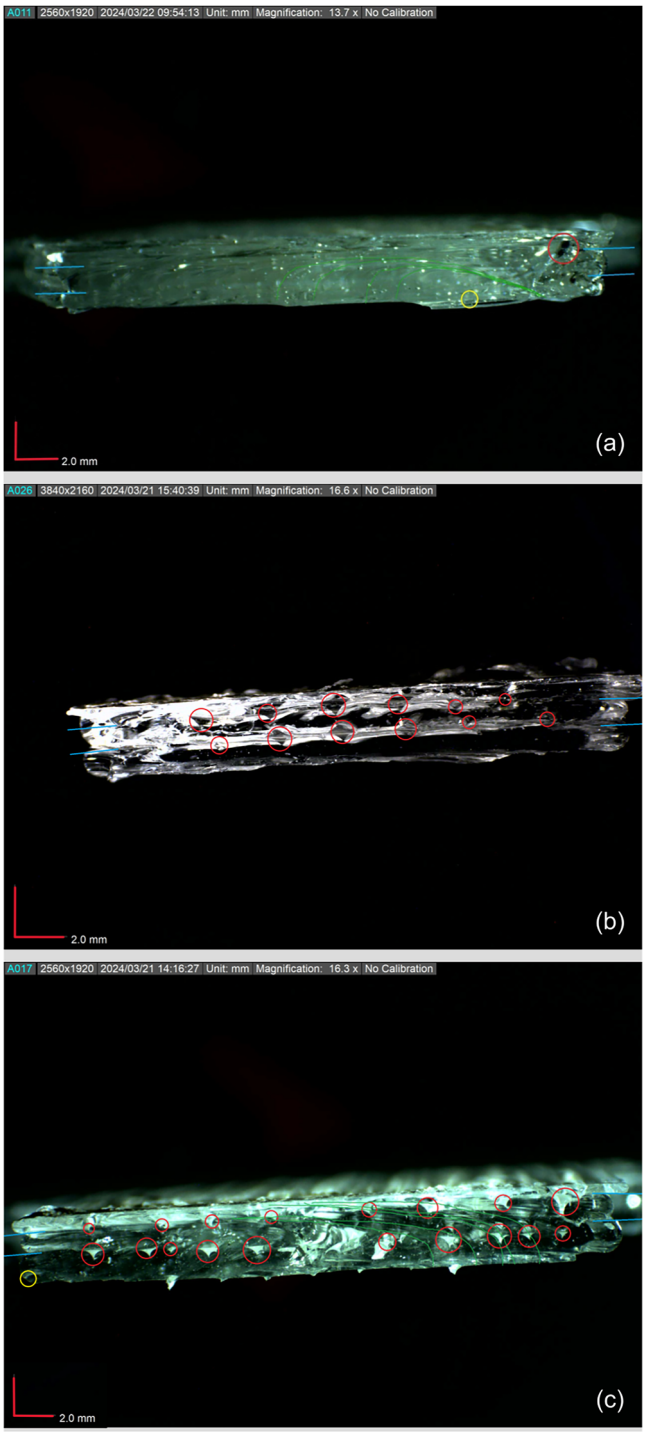


Fig. 9 | Optical micrographs of post bend test areas. Micrographs are from specimens following experiment 1, of post bend test area, where **a** 90° series, **b** 45°series, **c** 0°series.

automated-pulling method was selected for this study. A Vitri-Glass™ instrument (see Fig. 10) was employed to manufacture semi-infinite glass rods, measuring 3.5 mm in diameter. This process is the same as that utilised to commonly fabricate glass filament (i.e., rods), for glass 3D printing. An advantage of using the Vitri-Glass system is the ability to select the exact glass composition required for 3D printing, thereby avoiding the mixing of different batches or types of glass and reducing the risk of residual stresses caused by mismatched thermal expansion. The rod tests additionally serve as a baseline for comparing between the 3D-printed specimens with a simplified geometry made from the same material.

The Vitri-Glass instrument comprises a heated chamber and a crucible, featuring concentric exit holes. Solid glass is initially input to the crucible, and converted to glass rods as it passes through the heated chamber into ambient conditions. In this study, soda-lime sheet glass "001100 Tekta Clear" from Bullseye-Glass Co., was employed as the glass material. The sheet glass was sourced as waste from a glass workshop (in the previously unfired condition). Glass was loaded into the crucible, before heating the chamber to 790 °C. A thermocouple was located at the chamber centre, to continuously monitor temperature. Holding at 790 °C temperature for 60 min allows the glass to achieve an optimal viscosity for the "pulling" process, with a calculated viscosity of $10^{7.9}$ Poise. Subsequently, glass rods with a diameter of 3.5 mm were drawn at a speed of 11.3 mm/s.

Positioned beneath the heated chamber is a motorised system, responsible for pulling the glass from the crucible at constant speed. The speed selected by the user dictates the diameter of the rod. Increasing the pulling speed results in thinner glass rods and vice versa. The glass rods are automatically cut to lengths of approximately one metre, with this process repeating until the crucible is depleted. A single session was adequate to produce enough glass rods required for all 15 of the 3D printed dog-bone samples, with additional material remaining for glass rod testing.

To ensure a consistent diameter across all rods, regular inspections using calipers were conducted. A general tolerance for the drawn glass rods was 3.5 ± 0.1 mm. The diameters of the rods are listed in the supplementary information (see Supplementary Table 1). One challenge in manufacturing glass rods with this method is creating a homogeneous glass structure without air bubbles. When sheet glass is used as the input material, air bubbles can form between the sheets during heating. If the air has no escape route, they will be drawn into the glass rod, introducing weak points that can affect the mechanical testing results. During the production of the 3.5 mm rods the inclusion of air bubbles could be avoided.

3D-Printing process

The glass rods produced also served as the feedstock for the 3D printed dog-bone samples. A Maple 4™ glass 3D printer (see Fig. 11) was employed to additively manufacture the samples. The printer operates as a cartesian FDM printer, with the nozzle manoeuvring along the XY-axis, while the printing platform moves vertically in the Z-direction. The printer's nozzle was heated to 940 °C, as measured by the thermocouple positioned in the tip of the nozzle. The nozzle extends into a heated chamber, maintained at 500 °C, slightly below the annealing temperature of the soda-lime silicate glass. The printing platform is a ceramic plate, referred to as a kiln shelf. Once at temperature, the glass rod can be extruded through the 1.5 mm diameter outlet of the nozzle.

Each sample was printed individually to ensure precision and consistency. A layer height and line width of 1 mm and 2 mm was selected, respectively. Due to the nature of the process, the initial layer is compressed into the platform to encourage bonding. This can result in the first layer measuring slightly under 1 mm. Once the first layer is complete, the printing platform moves down 1 mm, before depositing the next layer. The first layer speed was selected as 2 mm/s, while subsequent layers were printed at 4 mm/s. Upon conclusion of the printing process, each sample was annealed for 15 min at 500 °C, before passively cooling to room temperature in the insulated chamber. The glass separates from the ceramic plate during cooldown, as both materials contract at different rates. A release agent, known as kiln wash, is applied before printing, to ensure safe and easy separation. Although kiln wash residue adheres to the first layer of the glass specimens, it is not expected to influence the test results due to the chosen testing orientation. Additionally, out of the experience of the authors there will be no significant residual stress because of the subsequent annealing process. Residual stresses could not be monitored using optical devices, due to the intransparency of the 3D printed specimen.

Testing methods

The objective of these tests is to determine the strength of AM glass as a function of layer orientation. While this is typically assessed in AM

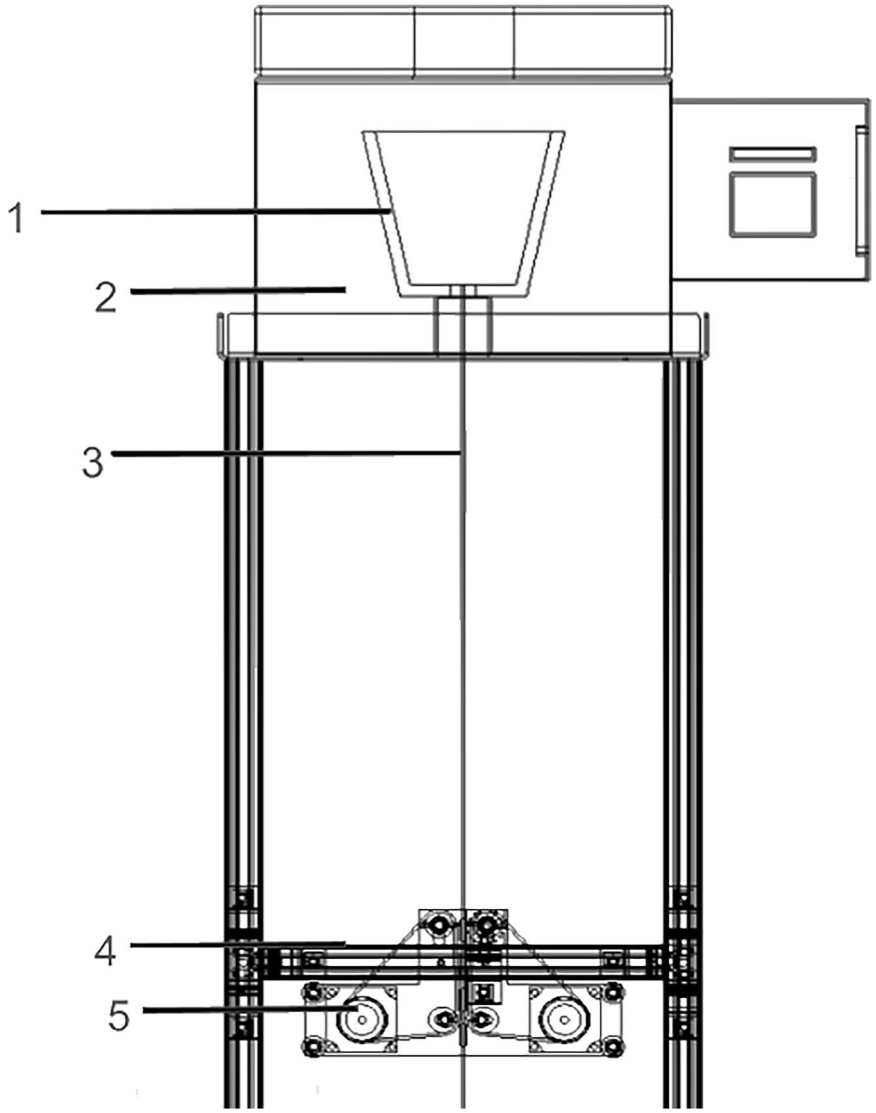


Fig. 10 | Vitri-Glass™ instrument schematic, where 1) crucible for input glass 2) heated chamber 3) drawn glass rod 4) stand and support structure 5) motorised pulling system.

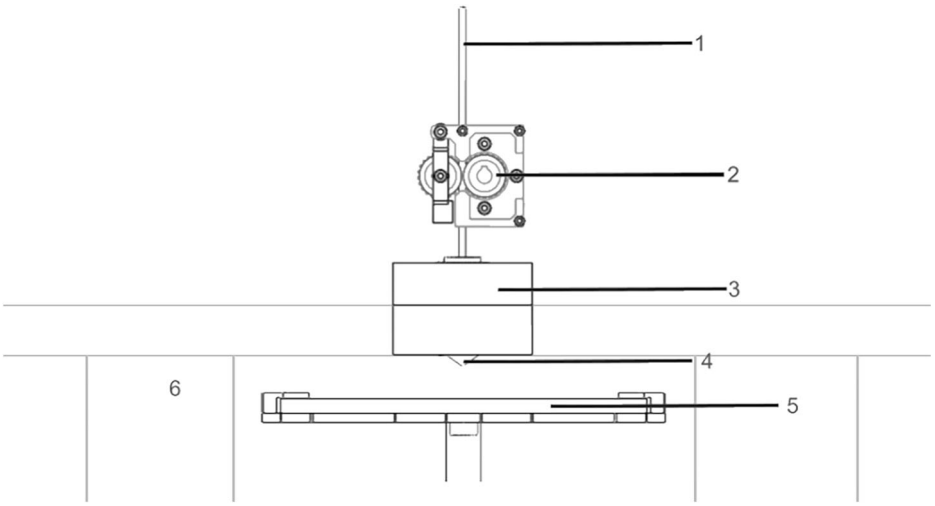


Fig. 11 | Glass 3D printer schematic, where 1) glass rod feedstock 2) direct drive extruder 3) heating block 4) nozzle tip 5) ceramic printing platform 6) heated chamber insulation.

(excluding glass) using a dog-bone tensile test[25], the brittleness of glass is considered here and a bending test is performed instead (tensile strength = bending tensile strength). Bending testing methods are typical for brittle materials (EN 843[19], EN 1288-3[18]).

A novel aspect of this study is the use of dog-bone test geometry in a bending setup, which is made possible by the design freedom of AM. The selected test geometry including the dimensions can be found in Fig. 6. The dog-bone geometry helps reduce stress concentrations at the support and load points due to the increased surface area, thereby promoting failure in the centre of the specimen's narrow section (see Fig. 4). The geometry was additionally validated using FEM analysis to ensure that no significant shear stresses occurred in the narrow section. A comparative value is necessary to evaluate the collected results. To establish this, individual thin glass rods are tested using the same four-point bending setup.

These bending tests offer a relatively straightforward and efficient method of mechanical evaluation, requiring minimal specimen preparation and a simple experimental setup. Established standards, such as EN 1288-3[18], are available to provide guidance and ensure consistency. However, it must be noted that for bending tests, specimens must exhibit high straightness and geometric uniformity, as deviations may compromise the accuracy and reliability of the results. Furthermore, elevated stress levels are often observed in bending tests at the sample edges, which may influence failure behaviour.

The method presented here can serve as a basis for the development of an established standard for determining the strength of additively manufactured glass structures. In the future, this will enable objective comparisons between different AM methods for glass.

Data Availability

The processed data supporting the findings of this study are available within the paper. Raw and/or unprocessed data required to reproduce any findings will be made available upon reasonable request.

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Author contributions

P.A.C., E.N., and D.V.: project administration, data curation, formal analysis, original draft, reviewing and editing. D.V.: production of the test samples, glass rods and AM components. P.A.C., E.N., and M.S.: conducted the 4-pt bend tests, microscopy and their analysis. M.S. and N.B.: project supervision, funding acquisition, writing, reviewing and editing. E.N. developed the concept for the bending tests. D.V., E.N., P.A.C., M.S., and N.B. have read and approved the manuscript.

Competing interests
The authors declare no competing interests.

Additional information
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Correspondence and requests for materials should be addressed to Dylan Vlahopoulos or Nick Birbilis.

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